California Polytechnic State University Pomona
DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING

Intro to Microelectronics Laboratory

ECE 2200L
Report #1

LAB 1 – Frequency Response of an RC Circuit

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Presented to
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January 29, 2025

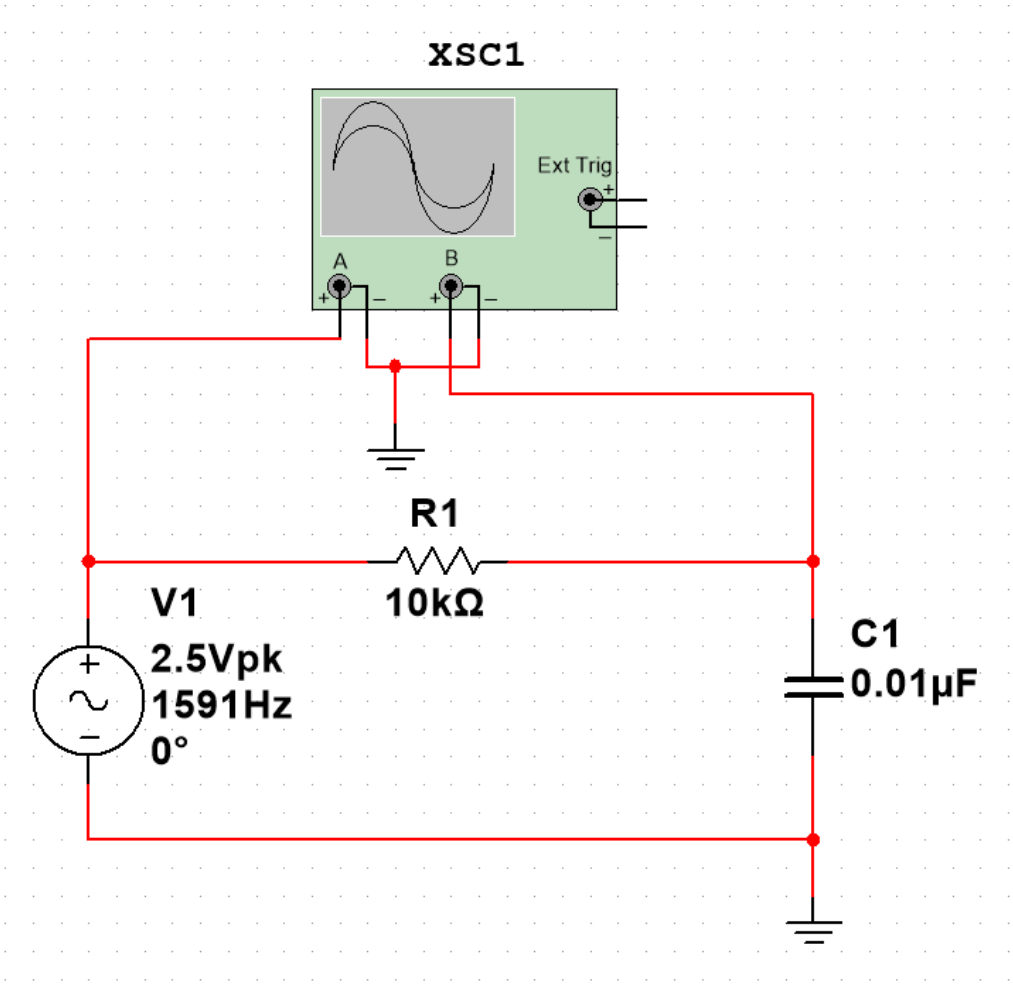
Introduction:

In this lab experiment our objective is to analyze the change in output voltage (Vout) and phase angle (θ) at different frequencies. We will use Excel and PSpice to determine our theoretical values. Then we will compare the theoretical values to our experimental values. At the end we will plot 20log(Vout/Vs) vs. phase angle using the experimental, theoretical, and simulation values. Our goal is to confirm the relation of frequency to output voltage and phase angle in an RC circuit.

Component List:

10kΩ resistor, 10μF capacitor, & 3 BNC to alligator cables

Circuit Schematic:



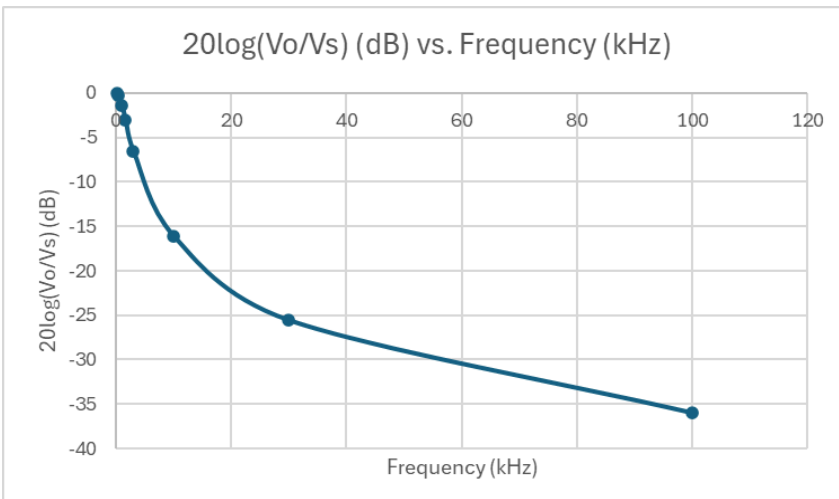
Pre-Lab:

Theoretical Data, Table 1:
Using 1/(2πRC), we found the center frequency fc = 1.591kHz. Our source voltage is 5 volts.

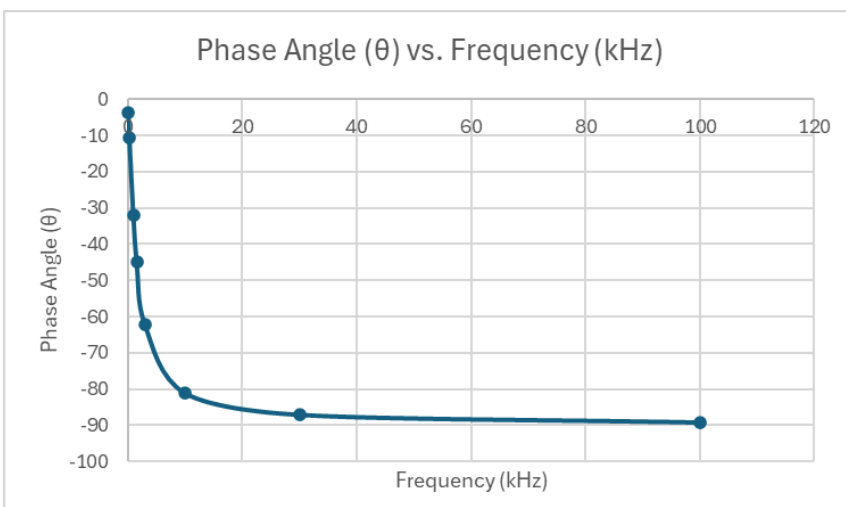
Frequency f(kHz)	0.1	0.3	1	1.591	3	10	30	100
Vout	4.99V	4.913V	4.24V	3.54V	2.343V	786mV	265mV	79.6mV
Vout/Vs	0.996 - j0.0627	0.966 - j0.1815	0.72 - j0.451	0.5 - j0.5	0.219 - j0.413	0.025 - j0.155	0.0028 - j0.053	0.00025 - j0.016
|Vo/Vi|	0.998	0.983	0.847	0.707	0.469	0.157	0.053	0.016
$20log(\frac{Vo}{Vi})$	-0.0174	-0.149	-1.44	-3.012	-6.58	-16.08	-25.514	-35.92
Phase angle (θ)	-3.6°	-10.7°	-32.15°	-45°	-62.06°	-80.96°	-86.96°	-89.09°

Theoretical Graphs

Graph 1A:



Graph 1B:

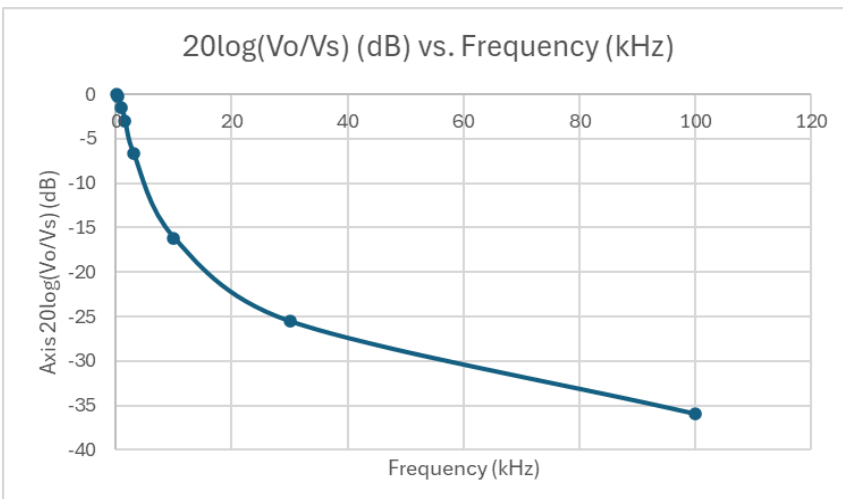


Lab Data:

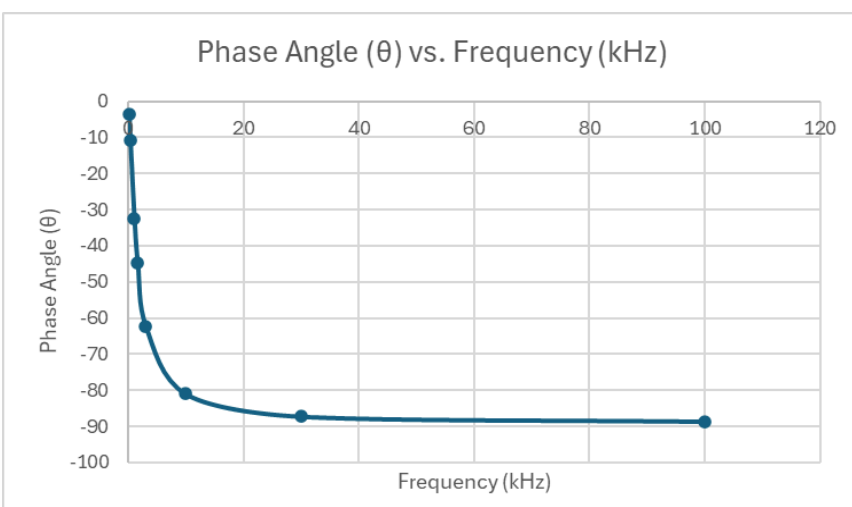
Simulation Data, Table 2:
Using 1/(2πRC), we found the center frequency fc = 1.591kHz. Our source voltage is 5Vpp, or 2.5Vpk.

Frequency f(kHz)	0.1	0.3	1	1.591	3	10	30	100
Vout (Vpp)	4.98V	4.9V	4.23V	3.53V	2.34V	780mV	264mV	79.3mV
$20log(\frac{Vo}{Vi})$	-0.035	-0.175	-1.45	-3.024	-6.6	-16.14	-25.55	-35.99
Phase angle (θ)	-3.55°	-10.65°	-32.42°	-44.87°	-62.28°	-81.05°	-87.22°	-88.63°

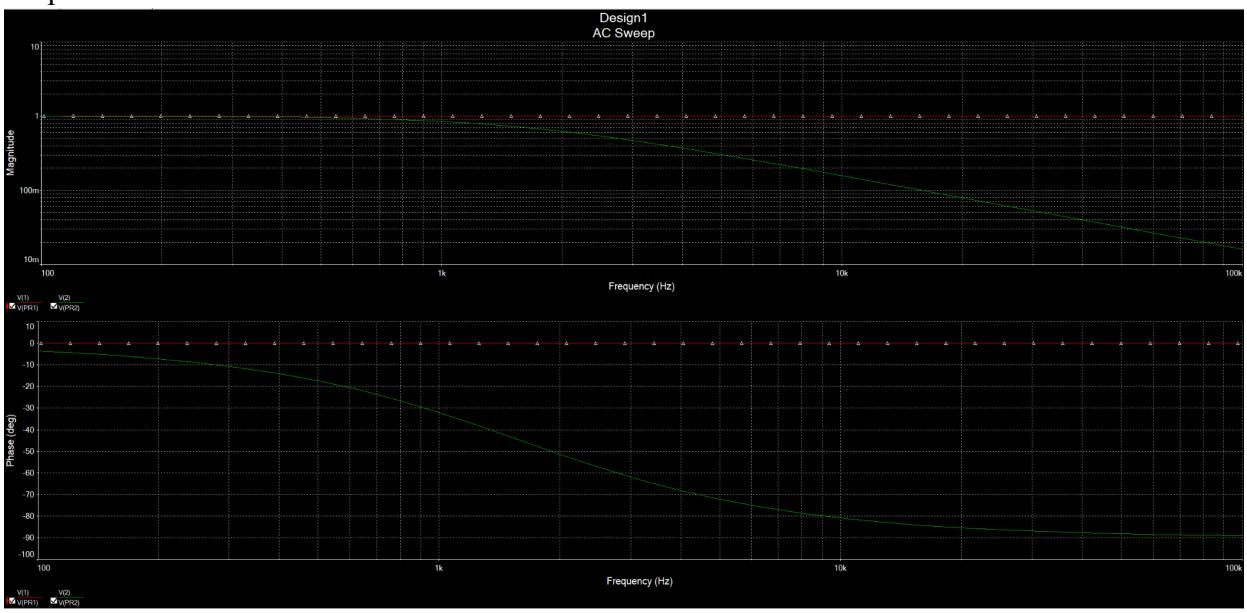
Simulation Graphs
Graph 2A:



Graph 2B:



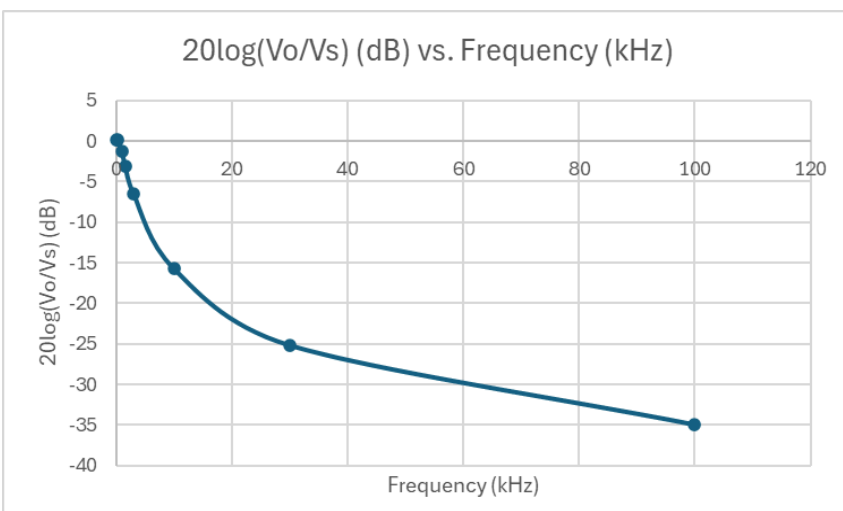
Graph 2C:



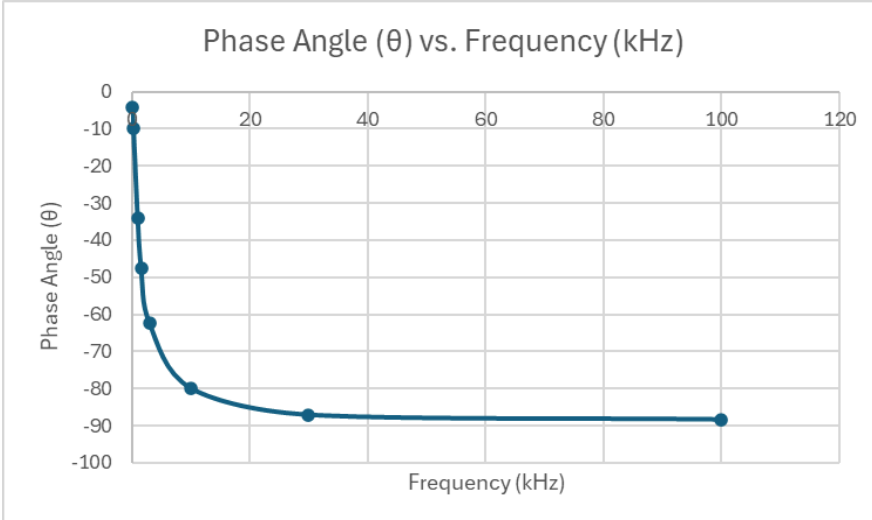
Experimental Data, Table 3:
Using 1/(2πRC), we found the center frequency fc = 1.591kHz. Our source voltage is 5 volts.

Frequency f(kHz)	0.1	0.3	1	1.591	3	10	30	100
Vout (Vpp)	5.1V	5.1V	4.3V	3.5V	2.37V	820mV	277mV	90mV
$20log(\frac{Vo}{Vi})$	0.172	0.172	-1.31	-3.1	-6.48	-15.7	-25.13	-34.89
Phase angle (θ)	-4.29°	-10.0°	-34.0°	-47.48°	-62.38°	-79.86°	-87.04°	-88.30°

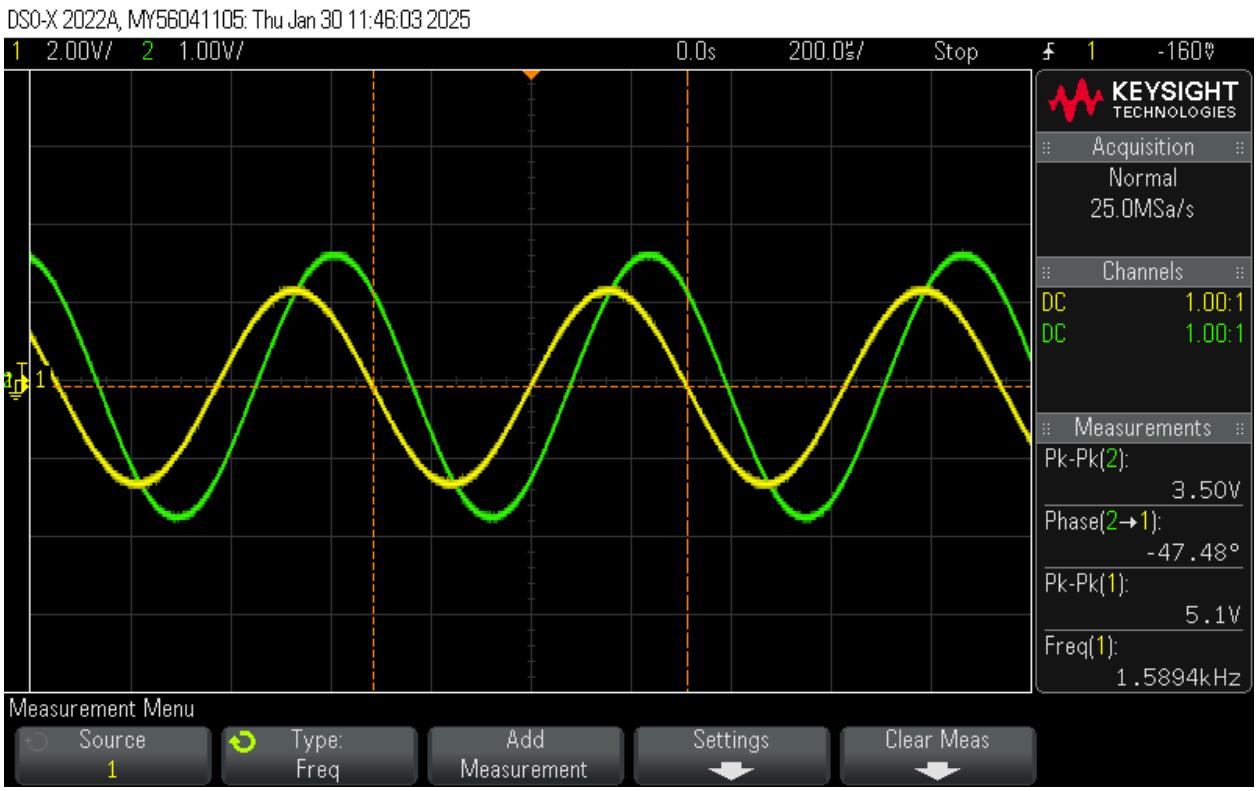
Experimental Graphs
Graph 3A:



Graph 3B:



Graph 3C:



Post-Lab:

When analyzing the theoretical data versus the simulation data we noted that the values are extremely similar. The slight discrepancies can be due to rounding or approximation errors. When comparing the experimental data to the theoretical/simulation we notice bigger discrepancies. This is most likely due to resistor and capacitor tolerances or other real world factors.

Conclusion:

After observing all of our data and graphs, we confirmed that all of our datasets follow the same trends. As frequency increases, the output voltage decreases, the phase angle approaches -90°, and the gain ($20log(\frac{Vo}{Vi})$) gets smaller and smaller. In conclusion, our data is very consistent and have consistent behavior, validating the RC circuit's low-pass filter characteristics.